

MOSFET – N-Channel, SUPREMOS

600 V, 76 A, 36 mΩ

FCH76N60N

Description

The SUPREMOS[®] MOSFET is ON Semiconductor's next generation of high voltage super-junction (SJ) technology employing a deep trench filling process that differentiates it from the conventional SJ MOSFETs. This advanced technology and precise process control provides lowest $R_{DS(on)}$ resistance, superior switching performance and ruggedness. SUPREMOS MOSFET is suitable for high frequency switching power converter applications such as PFC, server/telecom power, FPD TV power, ATX power and industrial power applications.

Features

- $R_{DS(on)} = 28 \text{ m}\Omega$ (Typ.) @ $V_{GS} = 10 \text{ V}$, $I_D = 38 \text{ A}$
- Ultra Low Gate Charge (Typ. $Q_g = 218 \text{ nC}$)
- Low Effective Output Capacitance (Typ. $C_{oss(eff.)} = 914 \text{ pF}$)
- 100% Avalanche Tested
- This Device is Pb-Free and is RoHS Compliant

Applications

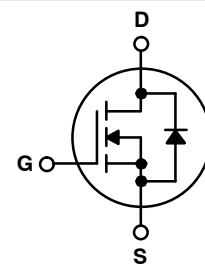
- Solar Inverter
- AC-DC Power Supply



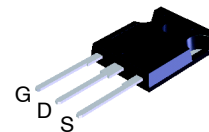
ON Semiconductor[®]

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V_{DS}	$R_{DS(on)}$ MAX	I_D MAX
600 V	36 mΩ @ 10 V	76 A

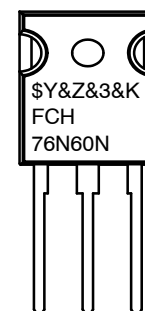


N-CHANNEL MOSFET



TO-247-3LD
CASE 340CK

MARKING DIAGRAM



\$Y	= ON Semiconductor Logo
&Z	= Assembly Plant Code
&3	= Numeric Date Code
&K	= Lot Code
FCH76N60N	= Specific Device Code

ORDERING INFORMATION

See detailed ordering and shipping information on page 2 of this data sheet.

FCH76N60N

MOSFET MAXIMUM RATINGS (T_C = 25°C unless otherwise noted)

Symbol	Parameter		FCH76N60N	Unit
V _{DSS}	Drain to Source Voltage		600	V
V _{GSS}	Gate to Source Voltage		±30	V
I _D	Drain Current	– Continuous (T _C = 25°C)	76	A
		– Continuous (T _C = 100°C)	48.1	
I _{DM}	Drain Current	– Pulsed (Note 1)	228	A
E _{AS}	Single Pulsed Avalanche Energy (Note 2)		8022	mJ
I _{AR}	Avalanche Current (Note 1)		25.3	A
E _{AR}	Repetitive Avalanche Energy (Note 1)		5.43	mJ
dv/dt	MOSFET dv/dt		100	V/ns
	Peak Diode Recovery dv/dt (Note 3)		20	
P _D	Power Dissipation	(T _C = 25°C)	543	W
		– Derate above 25°C	4.34	W/°C
T _J , T _{STG}	Operating and Storage Temperature Range		–55 to + 150	°C
T _L	Maximum Lead Temperature for Soldering, 1/8" from Case for 5 Second		300	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. Repetitive Rating: pulse-width limited by maximum junction temperature.
2. I_{AS} = 25.3 A, R_G = 25 Ω, starting T_J = 25 °C
3. I_{SD} ≤ 76 A, di/dt ≤ 200 A/μs, V_{DD} ≤ 380 V, starting T_J = 25 °C

THERMAL CHARACTERISTICS

Symbol	Parameter	FCH76N60N	Unit
R _{θJC}	Thermal Resistance, Junction to Case, Max.	0.23	°C/W
R _{θJA}	Thermal Resistance, Junction to Ambient, Max.	40	

PACKAGE MARKING AND ORDERING INFORMATION

Part Number	Top Mark	Package	Package Method	Reel Size	Tape Width	Quantity
FCH76N60N	FCH76N60N	TO-247-3LD	Tube	N/A	N/A	30 Units

FCH76N60N

ELECTRICAL CHARACTERISTICS (T_C = 25°C unless otherwise noted)

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit
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OFF CHARACTERISTICS

BV _{DSS}	Drain to Source Breakdown Voltage	I _D = 250 μ A, V _{GS} = 0 V, T _C = 25°C	600	–	–	V
$\Delta BV_{DSS} / \Delta T_J$	Breakdown Voltage Temperature Coefficient	I _D = 250 μ A, Referenced to 25°C	–	0.73	–	V/°C
I _{DSS}	Zero Gate Voltage Drain Current	V _{DS} = 480 V, V _{GS} = 0 V	–	–	10	μ A
		V _{DS} = 480 V, V _{GS} = 0 V, T _C = 125°C	–	–	100	
I _{GSS}	Gate to Body Leakage Current	V _{GS} = $\pm$ 30 V, V _{DS} = 0 V	–	–	$\pm$ 100	nA

ON CHARACTERISTICS

V _{GS(th)}	Gate Threshold Voltage	V _{GS} = V _{DS} , I _D = 250 μ A	2.0	–	4.0	V
R _{DS(on)}	Static Drain to Source On Resistance	V _{GS} = 10 V, I _D = 38 A	–	28	36	m Ω
g _{FS}	Forward Transconductance	V _{DS} = 20 V, I _D = 38 A	–	90	–	S

DYNAMIC CHARACTERISTICS

C _{iss}	Input Capacitance	V _{DS} = 100 V, V _{GS} = 0 V, f = 1 MHz	–	9310	12385	pF
C _{oss}	Output Capacitance		–	370	495	pF
C _{rss}	Reverse Transfer Capacitance		–	3.1	5	pF
C _{oss}	Output Capacitance	V _{DS} = 380 V, V _{GS} = 0 V, f = 1 MHz	–	195	–	pF
C _{oss(eff.)}	Effective Output Capacitance	V _{DS} = 0 V to 380 V, V _{GS} = 0 V	–	914	–	pF
Q _{g(tot)}	Total Gate Charge at 10 V	V _{DS} = 380 V, I _D = 38 A, V _{GS} = 10 V (Note 4)	–	218	285	nC
Q _{gs}	Gate to Source Gate Charge		–	39	–	nC
Q _{gd}	Gate to Drain “Miller” Charge		–	66	–	nC
ESR	Equivalent Series Resistance (G–S)	f = 1 MHz	–	1.0	–	Ω

SWITCHING CHARACTERISTICS

t _{d(on)}	Turn-On Delay Time	V _{DD} = 380 V, I _D = 38 A, R _G = 25 Ω (Note 4)	–	34	78	ns
t _r	Turn-On Rise Time		–	24	58	ns
t _{d(off)}	Turn-Off Delay Time		–	235	480	ns
t _f	Turn-Off Fall Time		–	32	74	ns

DRAIN-SOURCE DIODE CHARACTERISTICS

I _S	Maximum Continuous Drain to Source Diode Forward Current		–	–	76	A
I _{SM}	Maximum Pulsed Drain to Source Diode Forward Current		–	–	228	A
V _{SD}	Drain to Source Diode Forward Voltage	V _{GS} = 0 V, I _{SD} = 38 A	–	–	1.2	V
t _{rr}	Reverse Recovery Time	V _{GS} = 0 V, I _{SD} = 38 A, dI _F /dt = 100 A/ μ s	–	612	–	ns
Q _{rr}	Reverse Recovery Charge		–	16	–	μ C

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

4. Essentially Independent of Operating Temperature Typical Characteristics.

TYPICAL CHARACTERISTICS

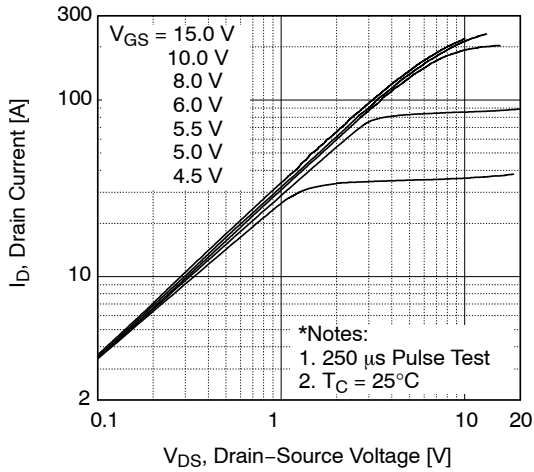


Figure 1. On-Region Characteristics

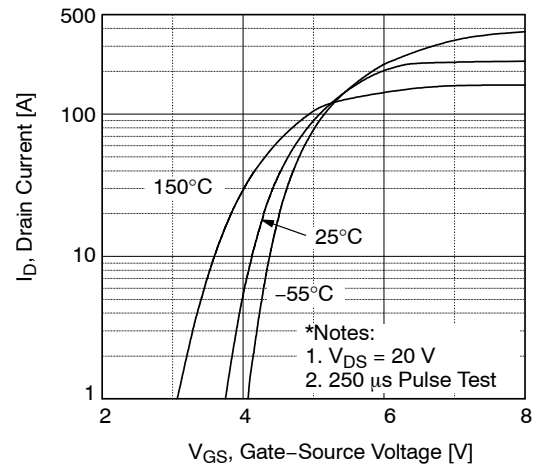


Figure 2. Transfer Characteristics

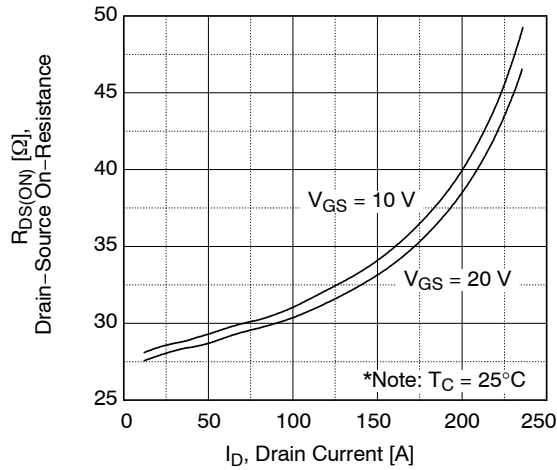


Figure 3. On-Resistance Variation vs. Drain Current and Gate Voltage

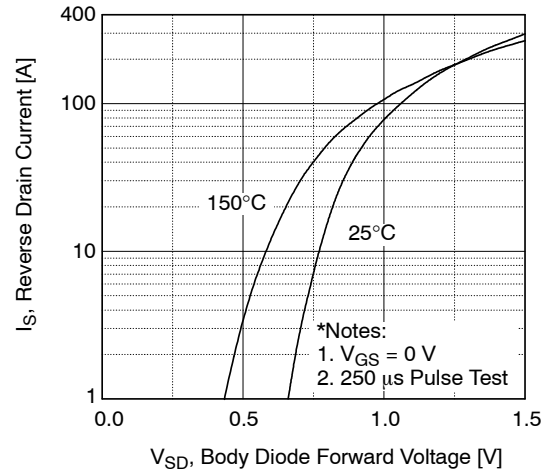


Figure 4. Body Diode Forward Voltage Variation vs. Source Current and Temperature

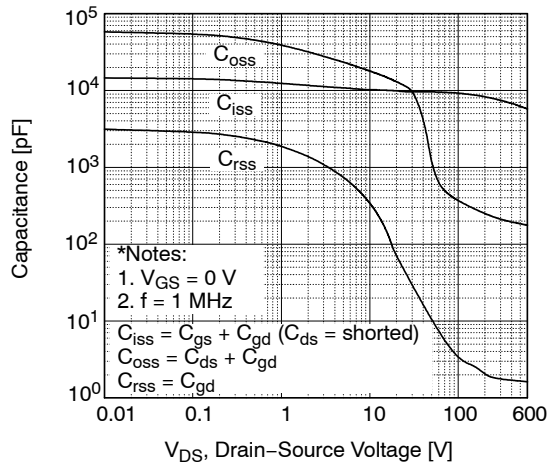


Figure 5. Capacitance Characteristics

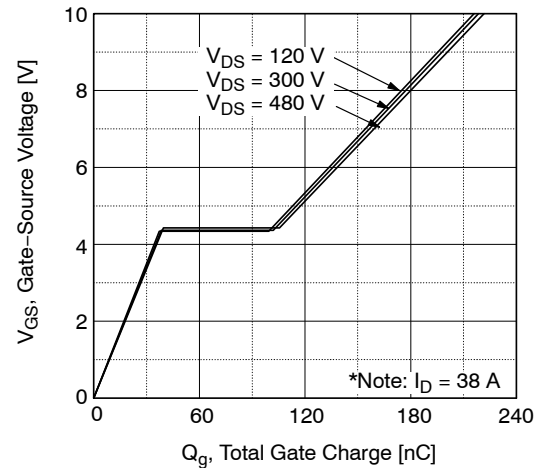


Figure 6. Gate Charge Characteristics

TYPICAL CHARACTERISTICS (continued)

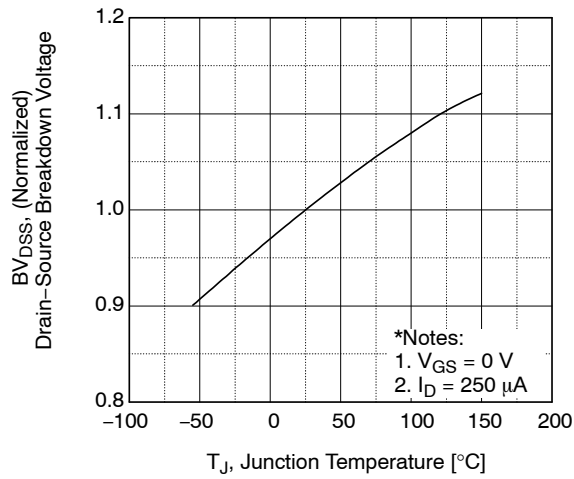


Figure 7. Breakdown Voltage Variation vs. Temperature

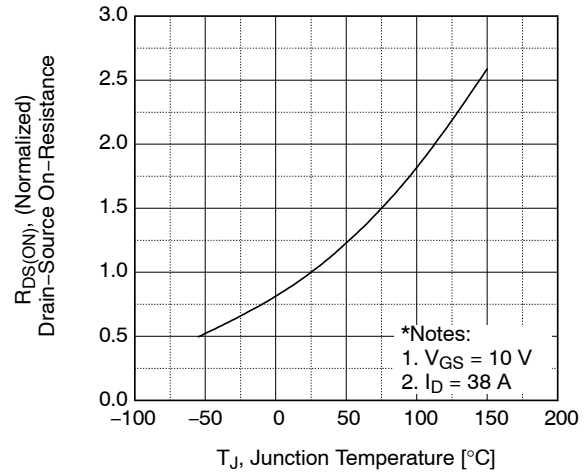


Figure 8. On-Resistance Variation vs. Temperature

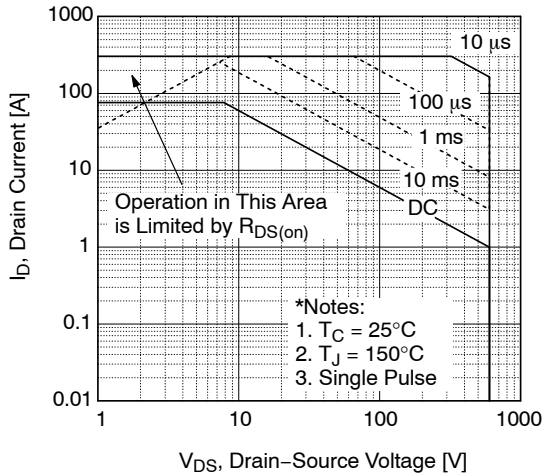


Figure 9. Maximum Safe Operating Area

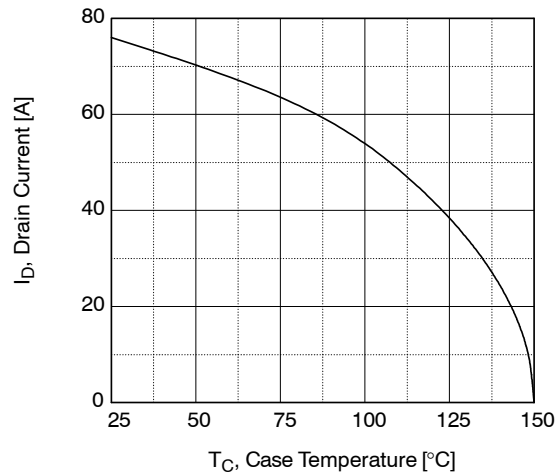


Figure 10. Maximum Drain Current vs. Case Temperature

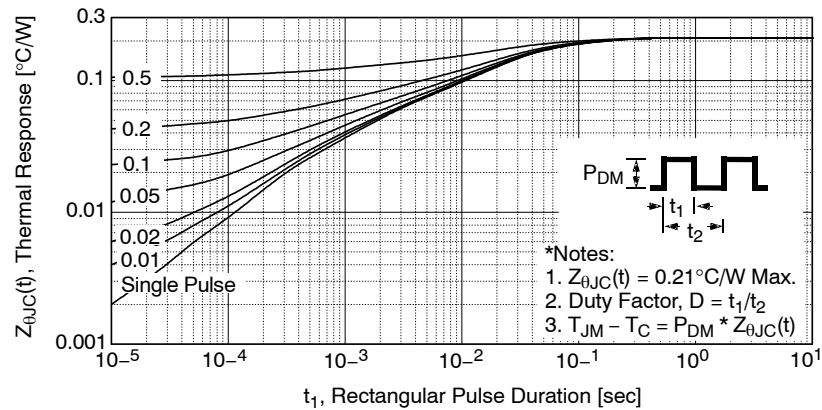


Figure 11. Transient Thermal Response Curve

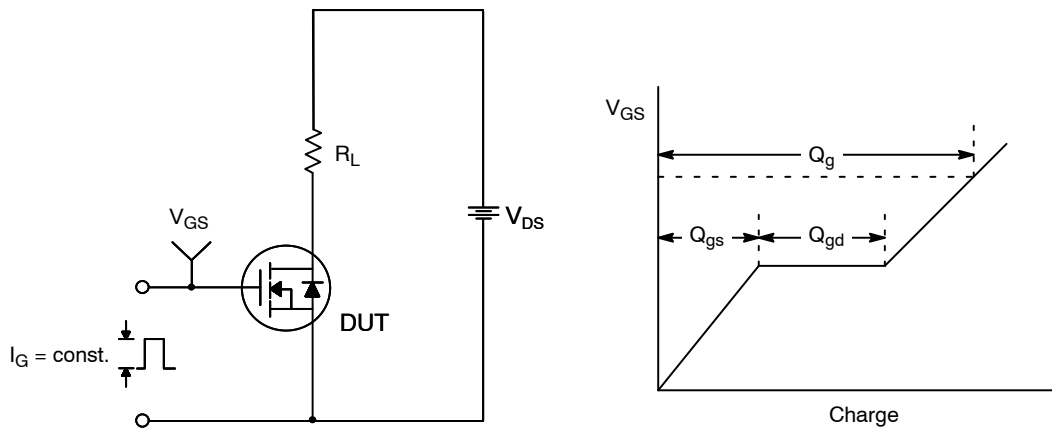


Figure 12. Gate Charge Test Circuit & Waveform

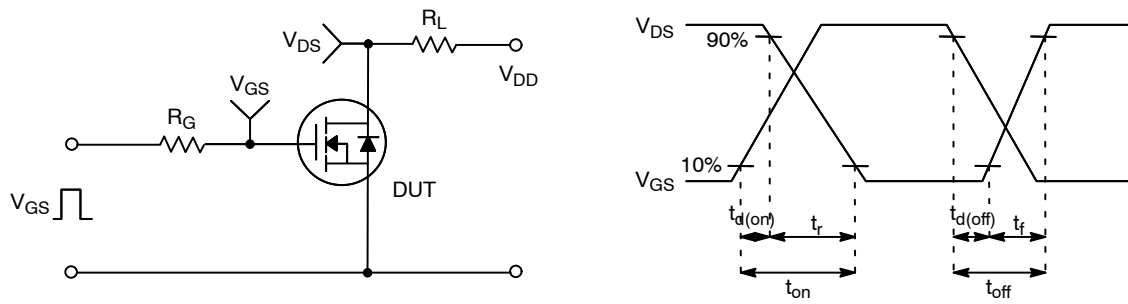


Figure 13. Resistive Switching Test Circuit & Waveforms

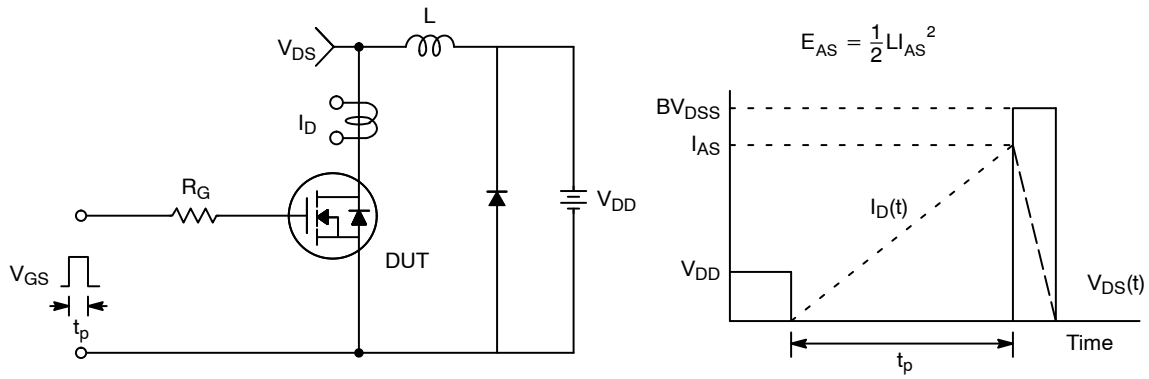


Figure 14. Unclamped Inductive Switching Test Circuit & Waveforms

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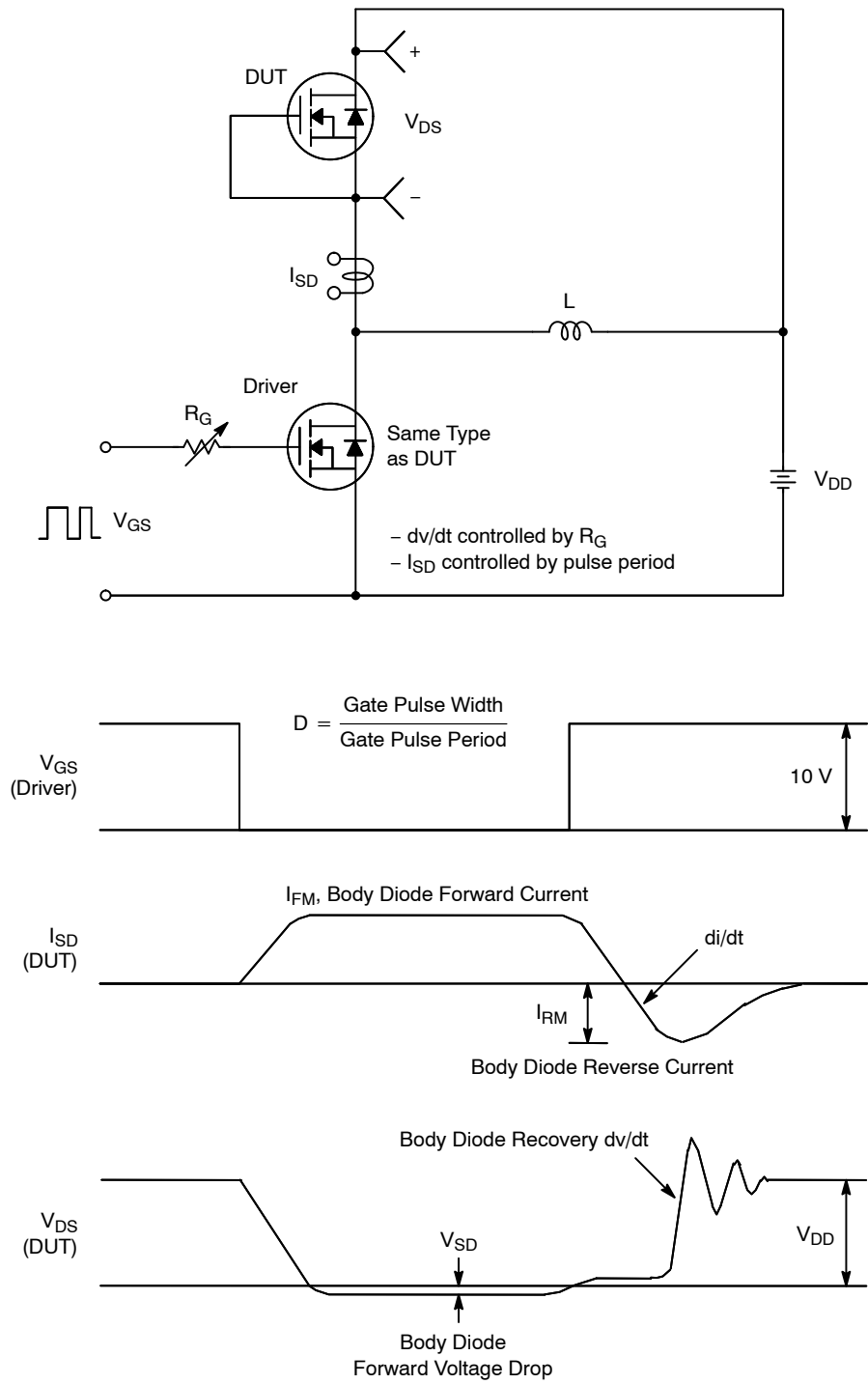
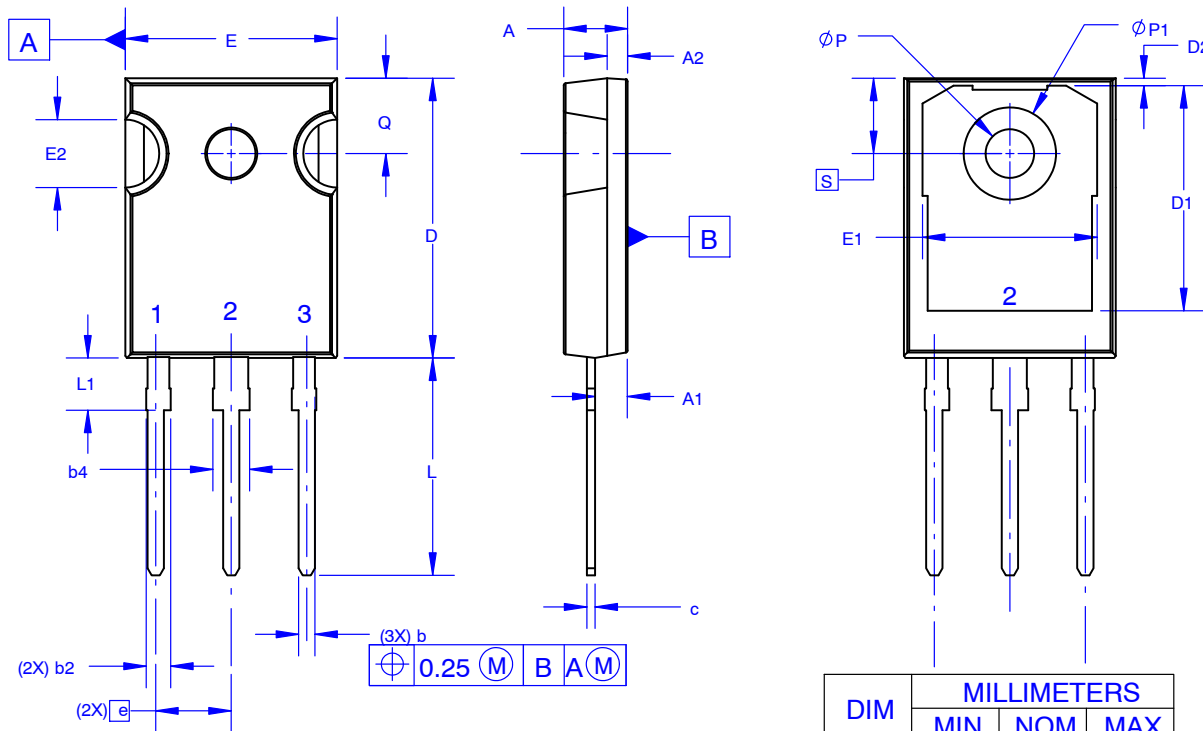


Figure 15. Peak Diode Recovery dv/dt Test Circuit & Waveforms

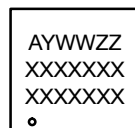
TO-247-3LD SHORT LEAD
CASE 340CK
ISSUE A

DATE 31 JAN 2019



NOTES: UNLESS OTHERWISE SPECIFIED.

- A. DIMENSIONS ARE EXCLUSIVE OF BURRS, MOLD FLASH, AND TIE BAR EXTRUSIONS.
B. ALL DIMENSIONS ARE IN MILLIMETERS.
C. DRAWING CONFORMS TO ASME Y14.5 - 2009.
D. DIMENSION A1 TO BE MEASURED IN THE REGION DEFINED BY L1.
E. LEAD FINISH IS UNCONTROLLED IN THE REGION DEFINED BY L1.

GENERIC
MARKING DIAGRAM*


XXXX = Specific Device Code
A = Assembly Location
Y = Year
WW = Work Week
ZZ = Assembly Lot Code

*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "•", may or may not be present. Some products may not follow the Generic Marking.

DIM	MILLIMETERS		
	MIN	NOM	MAX
A	4.58	4.70	4.82
A1	2.20	2.40	2.60
A2	1.40	1.50	1.60
b	1.17	1.26	1.35
b2	1.53	1.65	1.77
b4	2.42	2.54	2.66
c	0.51	0.61	0.71
D	20.32	20.57	20.82
D1	13.08	~	~
D2	0.51	0.93	1.35
E	15.37	15.62	15.87
E1	12.81	~	~
E2	4.96	5.08	5.20
e	~	5.56	~
L	15.75	16.00	16.25
L1	3.69	3.81	3.93
ϕP	3.51	3.58	3.65
$\phi P1$	6.60	6.80	7.00
Q	5.34	5.46	5.58
S	5.34	5.46	5.58

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